

DHEERAJ PRAKASH S

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EXECUTIVE SUMMARY

Full Stack Software Engineer with 3+ years of building high-performance, scalable backend systems and intelligent data workflows. Specialized in Python-based API development with Django and FastAPI, progressing from full-stack Django applications to AI-powered microservices using LLMs, RAG architectures, and AWS cloud infrastructure. Delivered enterprise platforms process 100K+ daily transactions with 95% latency reduction through async APIs, distributed architectures, and optimized database strategies.

TECHNICAL SKILLS

Backend: Python, FastAPI, Django/DRF, Microservices, Async API Design

AI/ML: LLMs, RAG, LangChain, LangGraph, NLP

Cloud & DevOps: AWS (EC2, S3, Lambda, SNS, SQS, DynamoDB), Docker, Kubernetes, CI/CD

Frontend: React, Next.js, TypeScript

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, OpenSearch, Query Optimization, Caching (Redis)

PROFESSIONAL EXPERIENCE

Software Developer

June 2024 – Present

Mallow Technologies

- Architected and deployed FastAPI-based microservices powering enterprise seller management and rebate processing systems handling 100,000+ daily transactions across multi-tenant AWS infrastructure (EC2, RDS PostgreSQL, S3)
- Built production-grade RAG (Retrieval Augmented Generation) architectures integrating LangChain, vector embeddings, and LLM orchestration to enable AI-powered document analysis, reducing manual data processing time by 90% and enabling contextual knowledge retrieval across 50,000+ documents
- Designed event-driven distributed systems using AWS SNS and SQS for asynchronous data ingestion and background processing, achieving 99.9% message delivery reliability and decoupling 12+ microservices for independent scaling
- Reduced API response latency by 85% through async FastAPI implementation, PostgreSQL query optimization with strategic indexing, Redis caching layers, and connection pooling strategies across high-volume endpoints serving 500+ requests/second
- Developed AI-powered CSV transformation pipeline using OpenAI GPT-4 API and LangChain, automatically converting unstructured vendor data uploads into system-ready formats with 95% accuracy, processing 10,000+ records daily
- Led development of Next.js + TypeScript administrative dashboard enabling vendors to manage products, incentive configurations, and analytics across distributed AWS infrastructure, improving frontend rendering performance by 90%

Junior Software Developer

June 2023 – May 2024

Mallow Technologies

- Built cloud storage platform using Django REST Framework with AWS S3 integration, implementing hierarchical folder structures, secure file upload/download workflows, and role-based access control serving 5,000+ users with 99.8% uptime
- Engineered corporate transport booking system with Django backend and React frontend, digitizing manual workflows and reducing operational coordination overhead by 85% through real-time booking management and automated driver allocation
- Optimized Django ORM query performance by 70% through strategic database indexing, select_related/prefetch_related usage, and PostgreSQL query analysis, reducing average API response time from 800ms to 240ms
- Developed multi-tenant SaaS architecture using Django with tenant-aware middleware and OpenSearch integration for high-performance filtering and analytics across 20+ enterprise clients
- Delivered React + Next.js administrative dashboards with dynamic configuration management, enabling business teams to control workflows and platform settings without engineering intervention

EDUCATION

Master of Business Administration – Finance

June 2023 – May 2025

Bharathidasan University, India

Bachelor of Engineering - Computer Science

June 2019 – May 2023

Saranathan College of Engineering, India

Projects

Azure Multimodal Compliance Orchestration Engine (Agentic RAG)

June 2026 – June 2026

- **Architected and deployed** an end-to-end production-grade automated Video Compliance QA pipeline using **LangGraph** for orchestration and **Azure OpenAI (GPT-4o)** for deterministic content auditing.
- **Implemented multimodal data ingestion** leveraging **Azure Video Indexer** to extract and process video transcripts and OCR, integrated with a RAG architecture using **Azure AI Search** for vector-based compliance rule retrieval.
- **Engineered observability and production monitoring** workflows by integrating **LangSmith** for granular LLM tracing and **Azure Application Insights** for real-time telemetry and full-stack performance logging.
- **Built a scalable API backend** using **FastAPI**, containerized for cloud deployment, transforming unstructured video inputs into actionable, structured JSON compliance reports.

Research Assistant - Django Multi-Agent AI Chatbot

August 2025 – August 2025

- Integrated a sophisticated multi-agent AI framework (Supervisor, Researcher, Analyst, Writer) into an existing Django application using LangGraph.
- Enhanced real-time chat interfaces, session management, and collaborative problem-solving capabilities.

Parkinson's Disease Progression Prediction

March 2023 – May 2023

- Refined ensemble machine learning models (Random Forest, Lasso, SVR, XGBoost, LGBM) for predicting Parkinson's disease progression.
- Optimized training pipelines on complex patient protein and clinical datasets containing UPDRS scores.

Passport Automation System

Nov 2021 – Nov 2021

- Upgraded a passport application and verification system using React and Django.
- Streamlined the three-step verification process and enhanced role-based authentication architectures using Django REST Framework.